



Season 2026/2027

Work Order – IFS09

Telemetry Receiver

Department: Electronics

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1. Executive Summary & Functional Description

1.1 Purpose

The purpose of this Work Order is to define the architectural specifications, hardware design requirements, manufacturing constraints, and validation test procedures for the IFS09 Telemetry Receiver Board. This custom PCB functions as the ground-station telemetry receiver responsible for capturing real-time vehicle dynamics, powertrain diagnostics, and battery management data transmitted by the IFS09 race car, and forwarding this telemetry stream to a ground station computer over USB-Serial.

1.2 System Overview & Operating Principle

The receiver board interfaces two primary active modules:

- **Processing & USB Interface Unit (Arduino Nano 3.0 with USB-C):** Employs an ATmega328P 8-bit microcontroller operating at 16 MHz with 5V logic. It manages SPI transactions with the wireless transceiver, unpacks telemetry payload frames, verifies data integrity, and streams raw telemetry packets to a ground PC over a high-speed UART-to-USB-C bridge (e.g., CH340X or FTDI / CP2102). It serves as the primary 5V power distribution node received via the USB-C bus.
- **Long-Range RF Transceiver (nRF24L01+ PA + LNA Module):** Operates in the 2.4 GHz ISM band using GFSK modulation with integrated Power Amplifier (PA) and Low-Noise Amplifier (LNA) for enhanced link margin and receiver sensitivity (up to -95 dBm @ 250 kbps). It is driven via hardware SPI (MOSI, MISO, SCK) with dedicated GPIOs for Chip Enable (CE) and Chip Select Not (CSN), with an optional Interrupt Request line (IRQ). It requires a clean, decoupled 3.3V power rail capable of supplying peak burst currents (up to ~115–150 mA) without causing brownout or packet drops.

2. System Architecture & Block Diagram

The diagram area below is reserved for the system architecture, power tree, and schematic block overview:

[SYSTEM ARCHITECTURE & BLOCK DIAGRAM]
(Reserved Area for Author Diagram)

3. Hardware & Design Requirements

3.1 Core Component Requirements

- **MCU Sub-module:** Footprint/header socket to mount a standard Arduino Nano 3.0 featuring a native USB-C connector.
- **RF Sub-module:** Standard 2×4 pin (2.54 mm pitch) socket header to mount the nRF24L01+ PA+LNA module.
- **Antenna Interface:** Chassis-mountable / edge-mount SMA Female (Jack) connector (50 Ω impedance) matched for an external 2.4 GHz omnidirectional whip antenna.
- **Power Regulation & Decoupling:** Dedicated on-board low-dropout regulator (LDO, 5V -> 3.3V, e.g., AMS1117-3.3 or AP2112K) to provide stable 3.3V to the nRF24L01+ module independently of the Arduino Nano internal 3.3V rail. Bulk decoupling: High-value electrolytic/tantalum capacitor (10 μ F – 47 μ F) and low-ESR ceramic capacitor (100 nF) placed in close proximity across the nRF24L01+ VCC and GND pins.

3.2 Signal Interfacing & Pinout Specification

Signal Name	Arduino Nano Pin	nRF24L01+ Pin	Function / Description
VCC_5V	5V / VIN	—	5V Bus derived from USB-C
VCC_3V3	3V3 (or LDO OUT)	VCC (Pin 2)	Clean 3.3V Power Supply
GND	GND	GND (Pin 1)	Common Ground Plane
CE	D7 (or D9)	CE (Pin 3)	Transceiver Chip Enable (RX/TX control)
CSN	D8 (or D10)	CSN (Pin 4)	SPI Chip Select Not (Active Low)
SCK	D13	SCK (Pin 5)	SPI Serial Clock
MOSI	D11	MOSI (Pin 6)	SPI Master Out Slave In
MISO	D12	MISO (Pin 7)	SPI Master In Slave Out
IRQ	D2 (INT0)	IRQ (Pin 8)	Optional Data Ready Interrupt

3.3 Testability Requirements (Test Points)

Dedicated surface mount / through-hole loop test points (min. 1.0 mm pad diameter, labeled on silkscreen) must be included for:

- **TP_5V:** USB 5.0 V input supply.
- **TP_3V3:** nRF24L01+ 3.3 V regulated supply rail.
- **TP_GND:** Primary circuit ground reference.
- **TP_SCK, TP_MOSI, TP_MISO:** SPI bus signals for oscilloscope/analyzer verification.
- **TP_CE, TP_CSN:** Radio control signals.
- **TP_UART_TX, TP_UART_RX:** Serial debugging lines.

3.4 Mechanical & Enclosure Constraints

- **Mounting Holes:** Four (4) symmetric mounting holes designed for M3 threaded heat-set inserts or screws (3.2 mm drill diameter, clearance for M3 bolt head/washer).
- **PCB Dimensions:** Compact rectangular form factor optimized to fit inside a custom 3D-printed enclosure (PLA/PETG).
- **Connector Access:** Edge clearance provided for (1) Direct, unblocked insertion of the USB-C cable into the Arduino Nano; (2) SMA connector extension protruding through the enclosure wall for external antenna attachment; and (3) Status indicator LEDs visible via enclosure light pipes or top surface.

4. Verification & Testing Requirements

Each manufactured board unit must undergo the following sequential testing protocol before sign-off and deployment:

4.1 Test Procedure 1: Power Supply Verification (Passive & Active)

- **Objective:** Ensure no short circuits exist and supply voltages remain within operational tolerances.
- **Tools:** Digital Multimeter (DMM), Bench Power Supply (Current Limited at 500 mA) / USB-C Supply.
- **Acceptance Criteria:** Resistance across TP_5V to TP_GND and TP_3V3 to TP_GND $> 10\text{ k}\Omega$ before power-up. $V(\text{TP_5V}) = 5.00\text{ V} \pm 0.25\text{ V}$. $V(\text{TP_3V3}) = 3.30\text{ V} \pm 0.10\text{ V}$ with peak ripple $< 30\text{ mVp-p}$ under RF transmission burst.

4.2 Test Procedure 2: MCU to Transceiver Interface Test

- **Objective:** Verify SPI bus integrity and successful initialization of the nRF24L01+ register map.
- **Tools:** Logic Analyzer / Oscilloscope, Arduino IDE / PlatformIO Serial Debugger.
- **Acceptance Criteria:** Execution of `radio.begin()` returns TRUE. `radio.isChipConnected()` validates presence of nRF24L01+. Clean SPI clock edge on TP_SCK without ringing or overshoot $> 0.5\text{ V}$.

4.3 Test Procedure 3: Serial Communication & PuTTY Data Transfer Test

- **Objective:** Confirm reliable, uncorrupted serial telemetry transfer to the base station PC.
- **Tools:** PC running PuTTY / Serial Terminal (configured at 115200 baud, 8-N-1, No Parity).
- **Acceptance Criteria:** Board connects over USB-C and enumerates virtual COM port without driver failure. PuTTY displays continuous formatted telemetry stream (JSON / CSV frame format) with 0% framing errors or dropped packets over a 10-minute continuous run.

4.4 Test Procedure 4: Operational Range & Car Link Test

- **Objective:** Validate RF link budget and packet reception rate (PRR) under track conditions.
- **Tools:** IFS09 telemetry transmitter on the vehicle / test rig, external 2.4 GHz antenna, ground PC.
- **Acceptance Criteria:** Packet Reception Rate (PRR) $\geq 95\%$ at line-of-sight distance $\geq 300\text{ m}$. Reliable link maintained throughout the entire FSAE/Formula Student autocross/endurance track perimeter under vehicle engine/inverter EMI conditions.

5. Test Execution Sign-Off Sheet

Test Item	Verification Method	Target Value / Criteria	Measured	Pass / Fail	Inspector	Date
1. 5V Supply Rail	DMM at TP_5V	4.75 V – 5.25 V				
2. 3.3V Supply Rail	DMM at TP_3V3	3.20 V – 3.40 V				
3. SPI Communication	Register Check via SW	isChipConnected() == true				
4. PuTTY Serial Stream	PuTTY @ 115200 Baud	0% errors / 10 min stream				
5. Field Range Test	Track test with car	PRR ≥ 95% @ ≥ 300 m				

6. Post-Design Requirement: Design Report

Upon completion of the PCB layout, hardware fabrication, and initial bench testing, a comprehensive Design Report document must be completed and submitted.

The Design Report must document:

- **Schematic & Layout Documentation:** Detailed schematic diagrams, layer stack-up, trace routing, and RF ground plane implementation.
- **Bill of Materials (BOM):** Complete list of manufacturer part numbers, exact SMD/through-hole footprints, and supplier sourcing.
- **3D Mechanical Integration:** Enclosure CAD integration, connector port clearances, and M3 insert stress considerations.
- **Validation Test Results:** Oscilloscope captures, measured voltage rails, serial logs, and vehicle link budget telemetry graphs recorded using Section 5.
- **Design Iterations & Recommendations:** Assembly issues encountered, thermal behavior, and proposed revisions for next-generation hardware.